EXP-2 Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

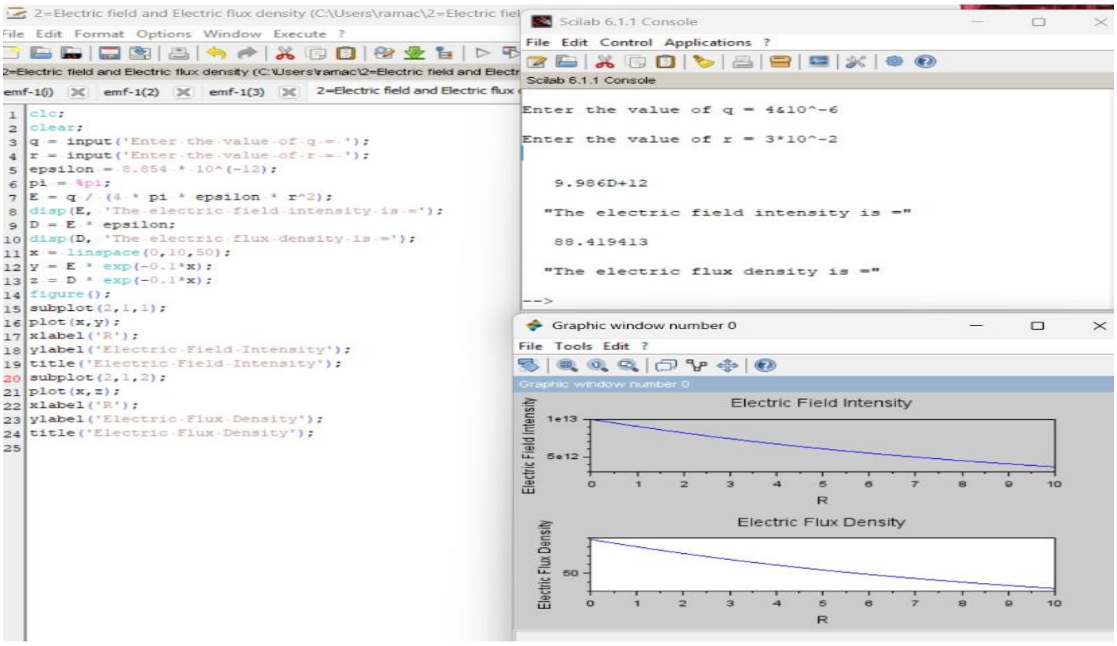
To write program for electric field and electric flux density in sphere using SCILAB.

Software required:

•SCILAB version 6.0.1

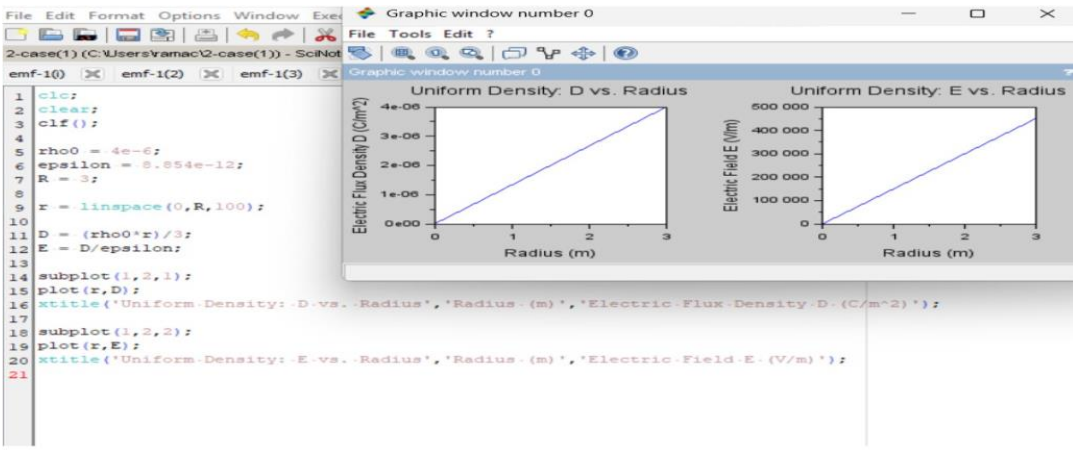
Practical output :

Test case -1



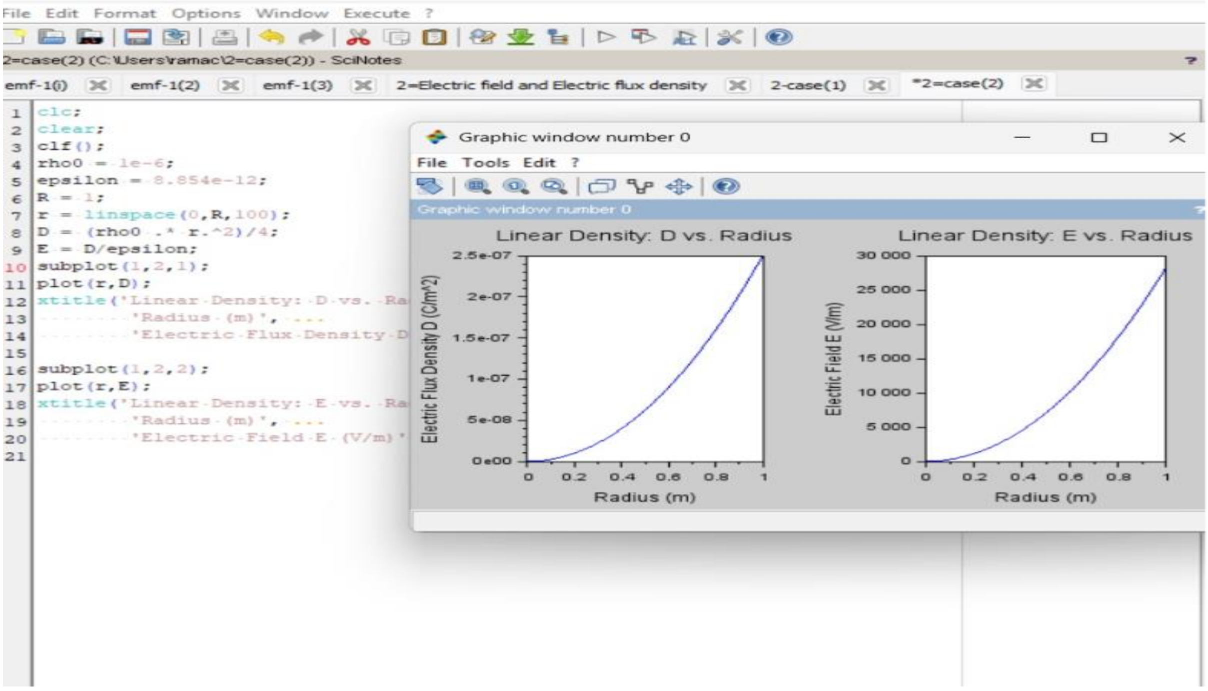
2-1

Test case-2 :



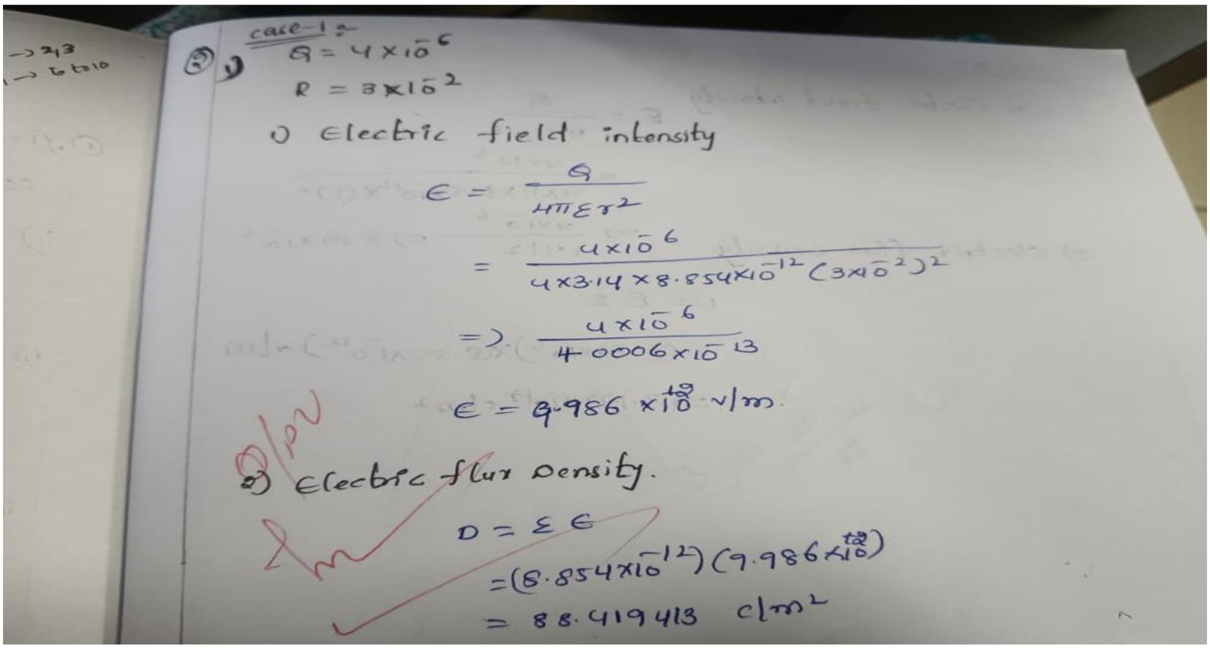
2-2

Test case-3:

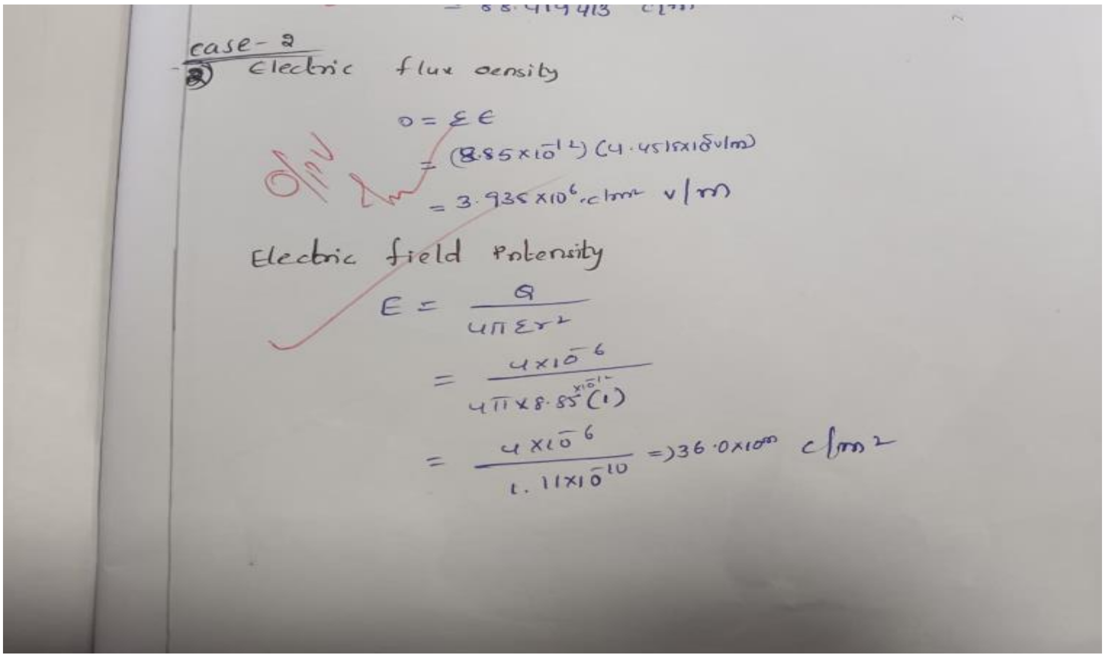


Theoretical output:

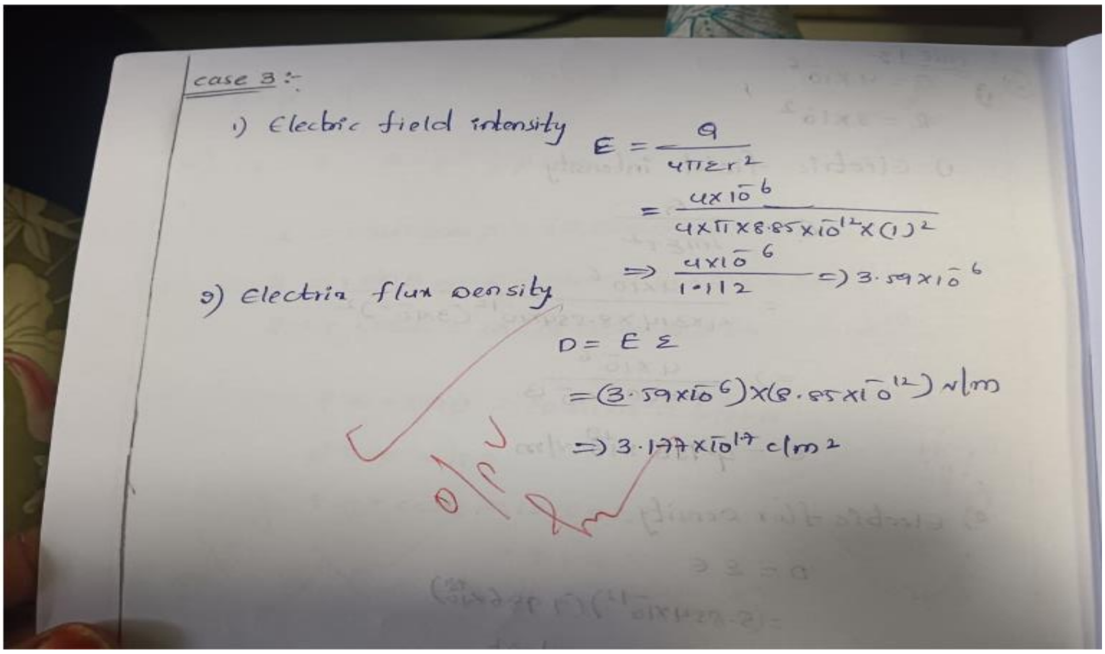
Test case -1



Test case-2:


case- 2
2) Electric flux density
$D = \varepsilon E$
$= (8.85\times10^{-12})(4.451\times10^{6} v/m)$
$= 3.935\times10^{-6} c/m^2$ v/m
Electric field intensity
$E = \frac{Q}{4\pi\varepsilon r^2}$
$= \frac{4\times10^{-6}}{4\pi\times8.85\times10^{-12}(1)}$
$= \frac{4\times10^{-6}}{1.11\times10^{-10}} \Rightarrow 36.0\times10^{m}$ c/m²

Test case-3:


case 3:-
1) Electric field intensity $E = \frac{Q}{4\pi\varepsilon r^2}$
$= \frac{4\times10^{-6}}{4\times\pi\times8.85\times10^{-12}\times(1)^2}$
$\Rightarrow \frac{4\times10^{-6}}{1.112} \Rightarrow 3.59\times10^{-6}$
2) Electric flux density
$D = E\,\varepsilon$
$= (3.59\times10^{-6})\times(8.85\times10^{-12})$ v/m
$\Rightarrow 3.177\times10^{-17}$ c/m²

Result:

Thus the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.